Sign Language Translation
for Indian Sign Language (ISL)

Aritra Ambudh Dutta (2530441) under the mentorship of Prof. Ashutosh Modi

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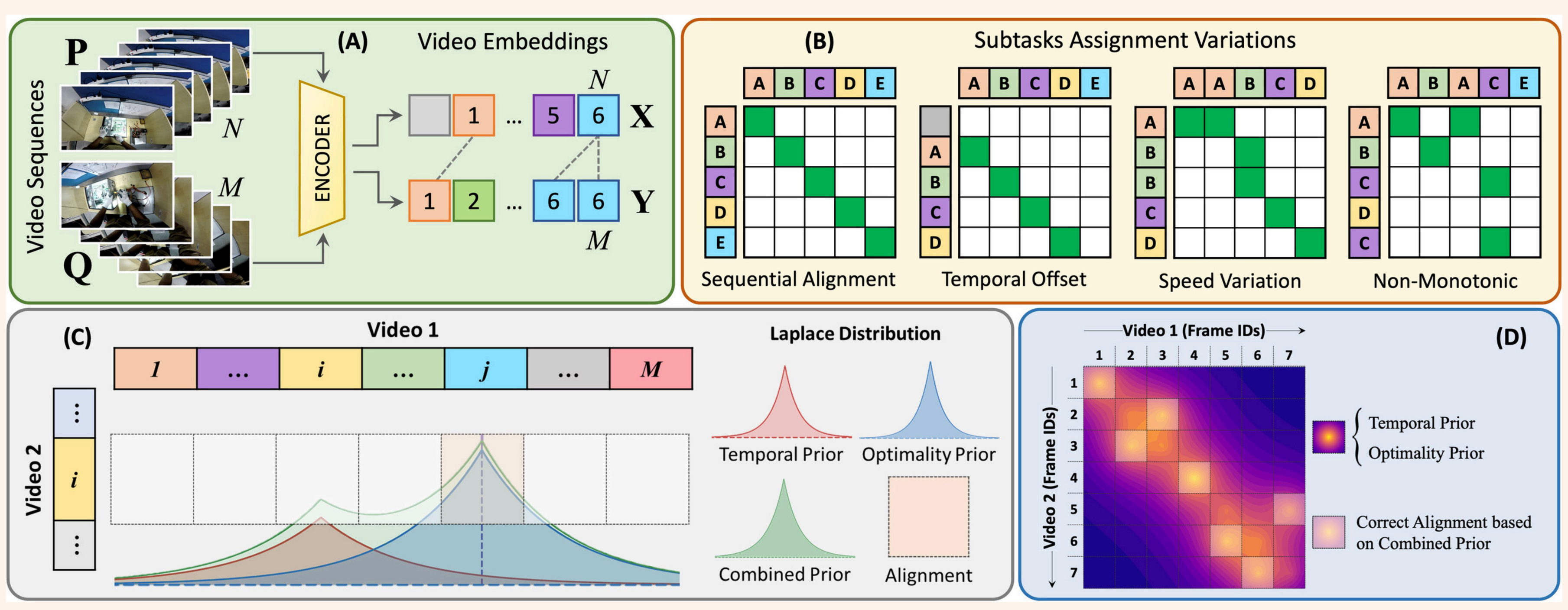
Abstract

This project addresses the challenge of translating Indian Sign Language (ISL) into spoken or written language at the word level by combining procedural learning and self-supervised visual modeling. We explore the integration of Optimal Transport Guided ProcedurE Learning (OPEL) and SHuBERT, a transformer-based self-supervised model for sign representation. OPEL enables flexible temporal alignment of gesture sequences by treating video frames as probability distributions and optimizing transport costs across sequences. SHuBERT, in contrast, learns contextualized embeddings from multi-stream visual input – specifically, body pose, hand, and face features. We also curated and preprocessed a large-scale ISL dataset comprising over 150,000 high-quality videos, the largest word-level ISL dataset to date. The overall goal is to build a multimodal gesture-to-text translation model that leverages both alignment flexibility and robust feature representation.



OPEL Framework Illustrated

(A) The encoder generates frame-wise embeddings from videos, facilitating subsequent OT calculations. (B) Pair-wise scenarios captured through the assignment matrix- from strictly synchronized actions to temporal shifts and differing action speeds, to non-monotonicity. (C) 1-D depiction of alignment of a single frame (i-th) of Video 2 with its best match frame (j-th) of Video 1, based on the proposed priors. (D) 2-D representation of the optimal alignment of frame sequences.

Introduction

Indian Sign Language (ISL), like other sign languages, conveys meaning through dynamic gestures involving hand shapes, orientations, facial expressions, and body movements. However, SLT (Sign Language Translation) systems face critical limitations: lack of labeled data, temporal variability in gestures, and the multimodal nature of sign expressions. Traditional models often assume monotonic alignment and suffer from rigid frame-wise matching. Our project proposes to tackle these challenges by leveraging two modern techniques:

OPEL: A procedure learning method that uses optimal transport to align frames from multiple gesture sequences. It accounts for temporal variability, background noise, and non-monotonicity in gesture videos.

SHuBERT: A BERT-style self-supervised transformer that learns contextual representations from video streams using masked cluster prediction on hand, face, and body pose features.

By applying these models to ISL, we aim to develop a robust translation system and make available a diverse ISL dataset to drive future research.

Research Objectives

- To build a multimodal model for Indian Sign Language word-level translation that combines alignment learning and visual representation learning.
- To implement and adapt the OPEL framework to ISL gesture sequences by modeling frame distributions using optimal transport, thereby enabling soft temporal alignment.
- To explore the application of SHuBERT's multi-stream transformer encoder to learn context-aware, self-supervised representations of ISL signs.
- To design and execute a scalable data pipeline for scraping, cleaning, and preprocessing ISL videos from public sources.
- To integrate OPEL and SHuBERT into a unified framework that improves generalization, interpretability, and performance of ISL translation.

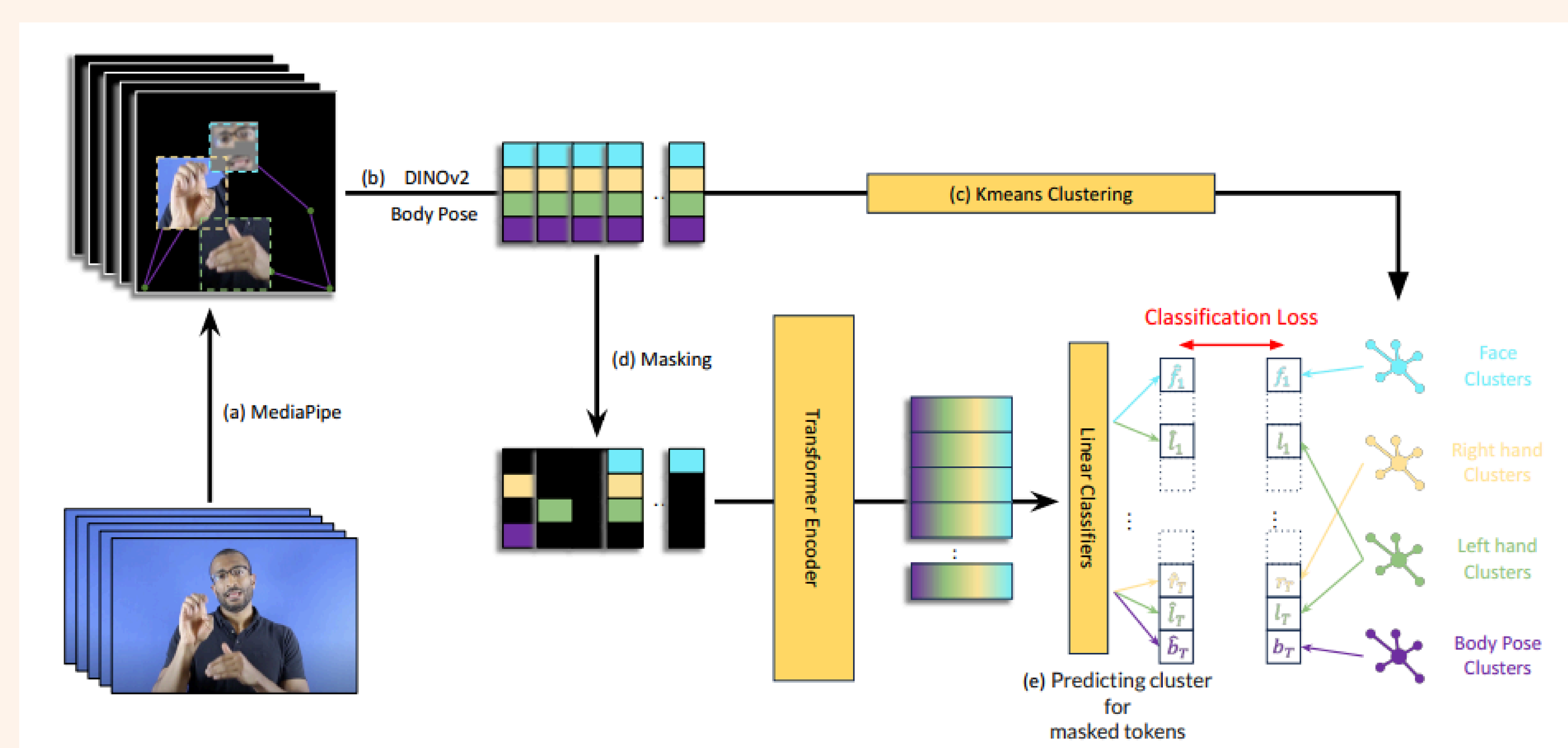
Study Methodology

1. Literature Review:
   - Conducted an in-depth review of Optimal Transport and the OPEL framework to understand soft alignment of video sequences.
   - Studied SHuBERT's masked prediction strategy across multiple visual streams.

2. Dataset Construction:
   - Collected ~150,000 ISL videos from public sources, covering diverse signers and settings.
   - Filtered and annotated clips to ensure clarity, linguistic relevance, and suitability for word-level modeling.

3. Pre-Processing Pipeline:
   - Applied YOLOv11 for person detection and MediaPipe for extracting hand, face, and body landmarks.
   - Used DINOv2 embeddings for high-resolution representation of hand and face image crops.

4. Model Implementation:
   - Adapted OPEL loss with Laplace-based priors and virtual frames to handle gesture variation and noise.
   - Handled background and unmatched frames using virtual nodes, improving robustness in variable signing conditions.



SHuBERT Pre-Training Pipeline

Multi-stream features (hands, face, and upper body pose) are extracted using MediaPipe and DINOv2, mapped to pre-computed clusters, partially masked, and used to train a transformer model to predict the masked cluster assignments.

5. Integration Strategy:
   - Currently integrating SHuBERT's multi-stream encoder into a modular OPEL-based pipeline that combines contextual representation learning with soft alignment.
   - Planned two-stage training: pretraining on existing sign corpora followed by ISL-specific fine-tuning.

Results

1. Dataset:
   - Curated and preprocessed ~150,000 high-quality ISL videos, making it the largest known word-level ISL dataset to date.

2. OPEL Implementation:
   - Successfully applied OPEL for ISL gesture alignment, incorporating temporal and optimality priors to handle varied sign durations and improve alignment robustness.

3. SHuBERT Integration :
   - Multi-stream features extracted; currently integrating SHuBERT's encoder into the OPEL pipeline for joint representation and alignment learning.

References

1. Chowdhury, S., Chandra, S., & Roy, K. (2024). OPEL: Optimal Transport Guided ProcedurE Learning. NeurIPS 2024.

2. Gueuwou, S., Du, X., Shakhnarovich, G., & Livescu, K. (2024). SHuBERT: Self-Supervised Sign Language Representation Learning via Multi-Stream Cluster Prediction. arXiv:2411.16765.